

Cancer Equity and Outcomes Analysis

Analysis of electronic health record data to identify causes of poor cancer outcomes among low-income patients, with recommendations to improve diagnosis outcomes, treatment access and survival for patients below the poverty line.

12K

Total Patients

720

Cancer Patients

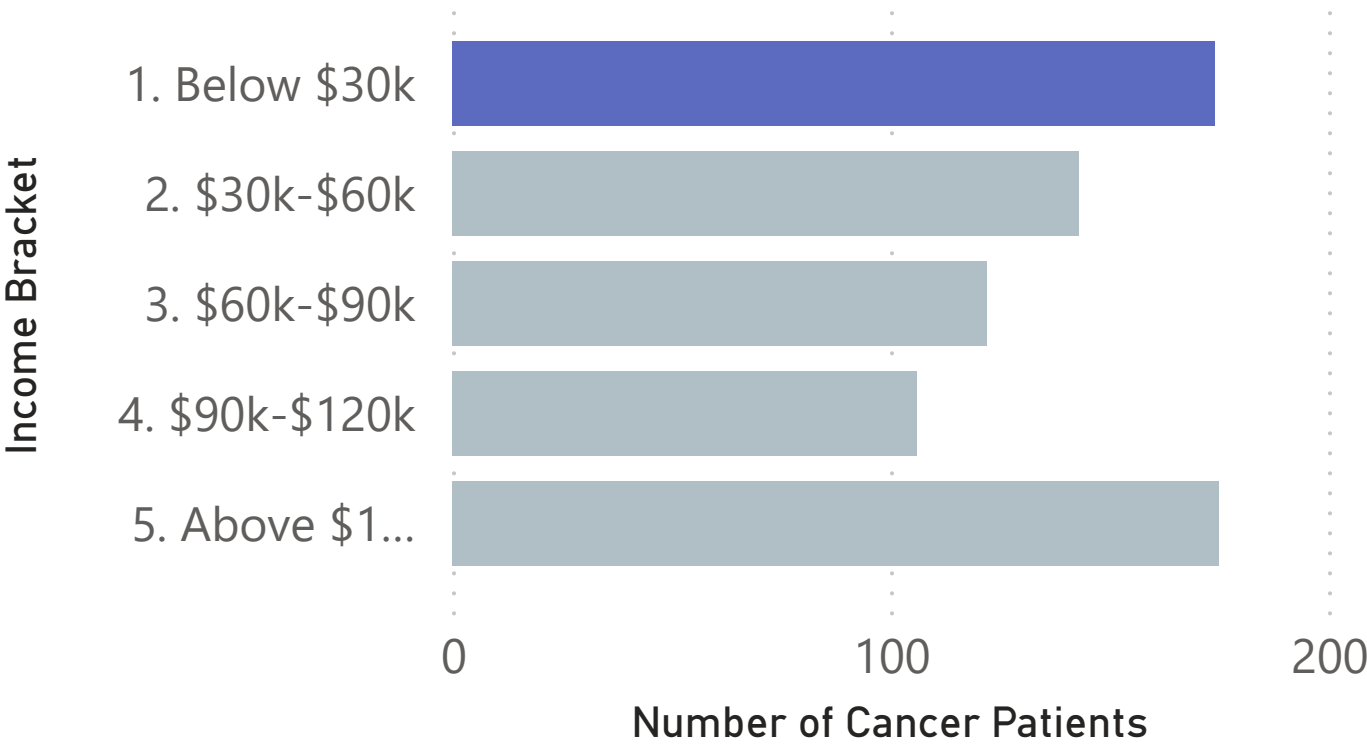
60.30

Avg Diagnosis Age

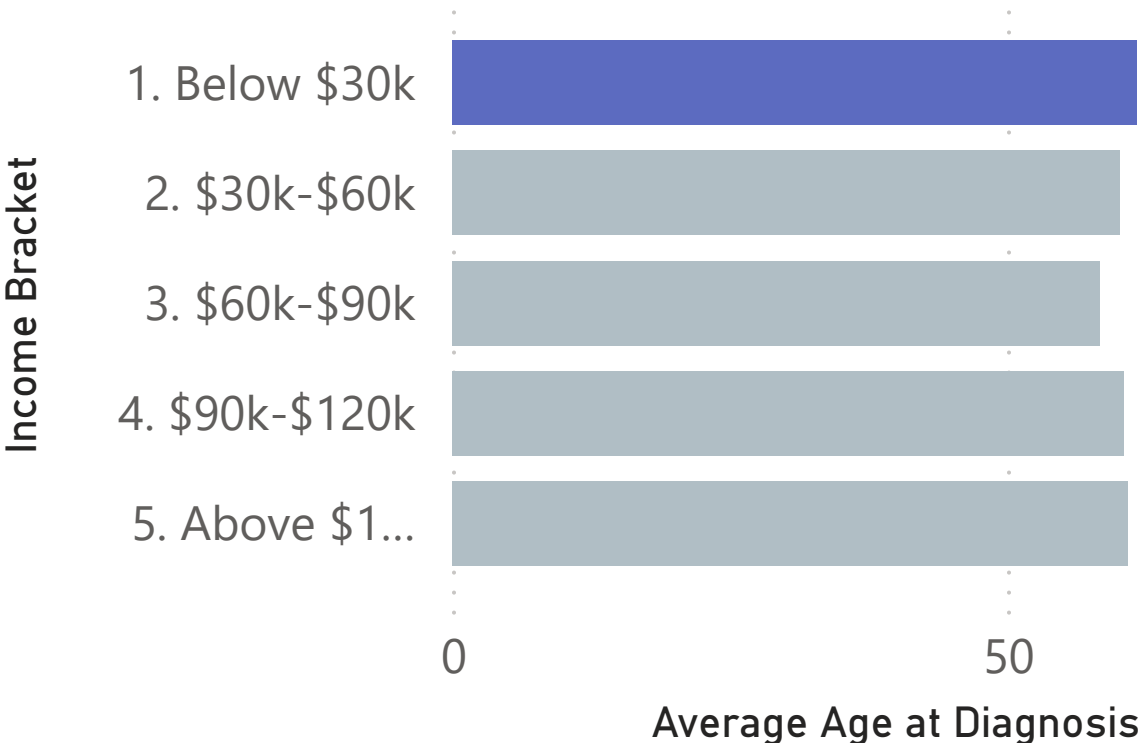
119.91K

Avg Out of Pocket Cost

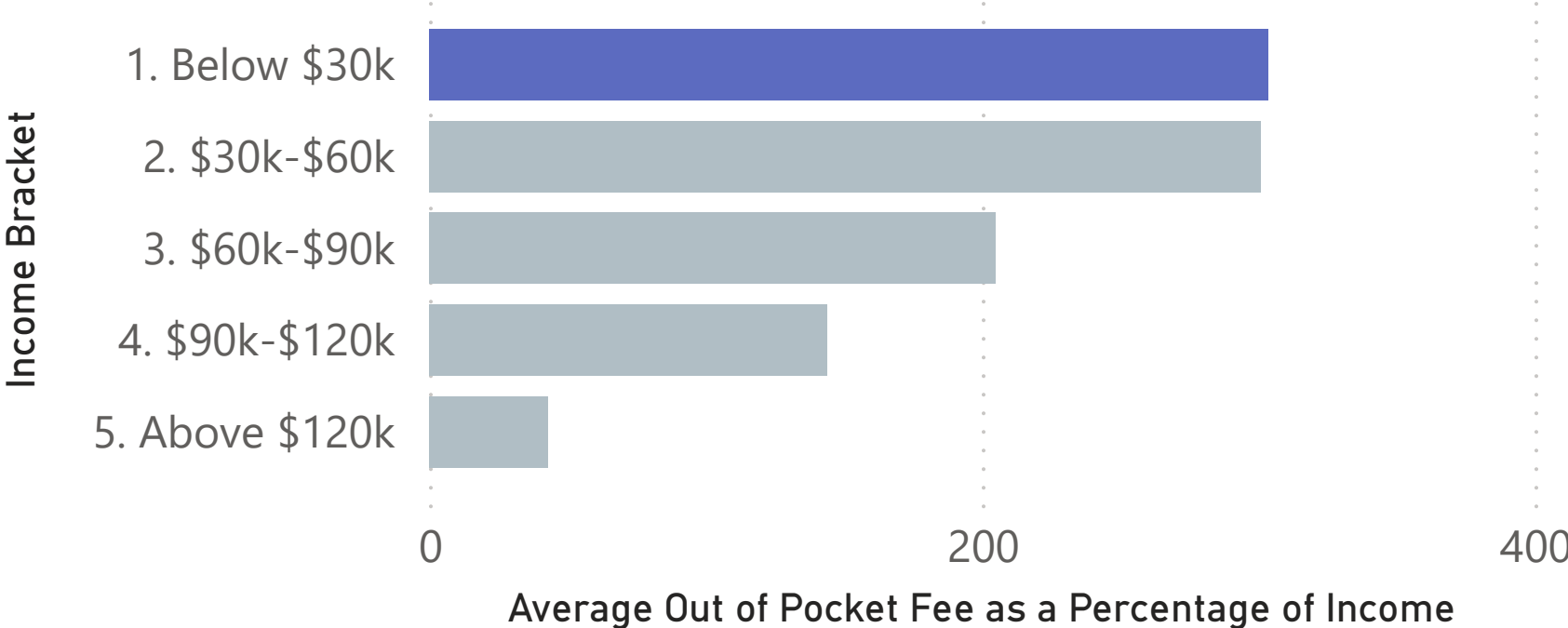
Cancer Patient Distribution by Income



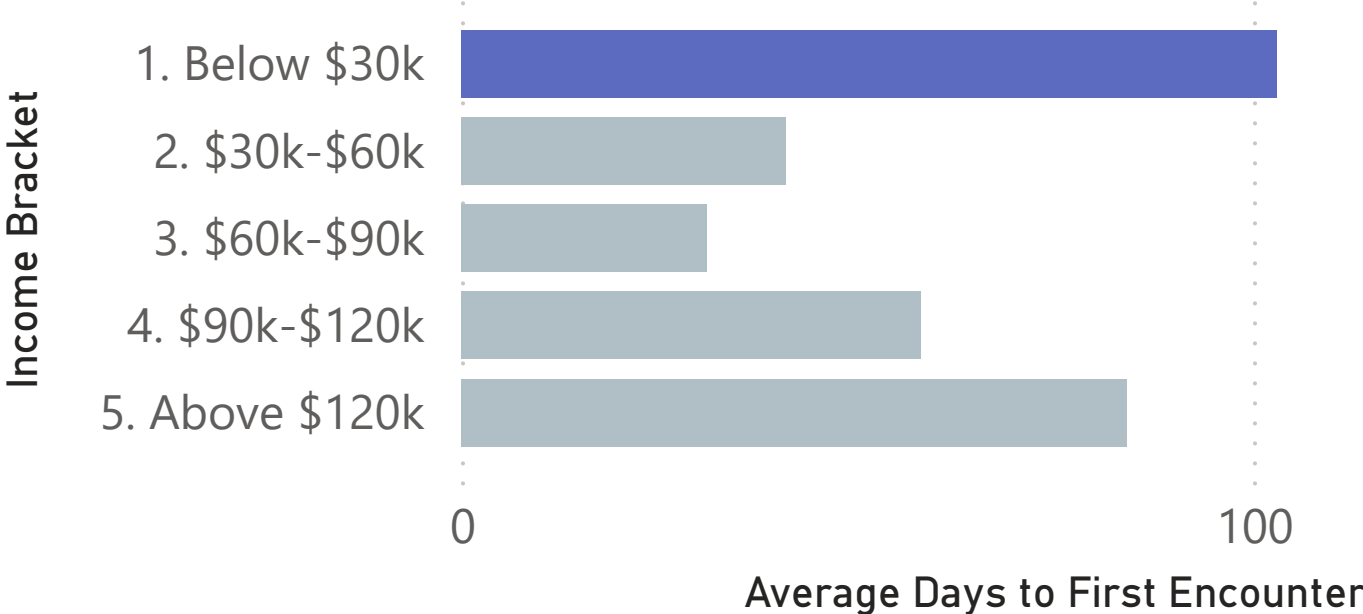
Age at Cancer Diagnosis by Income Bracket



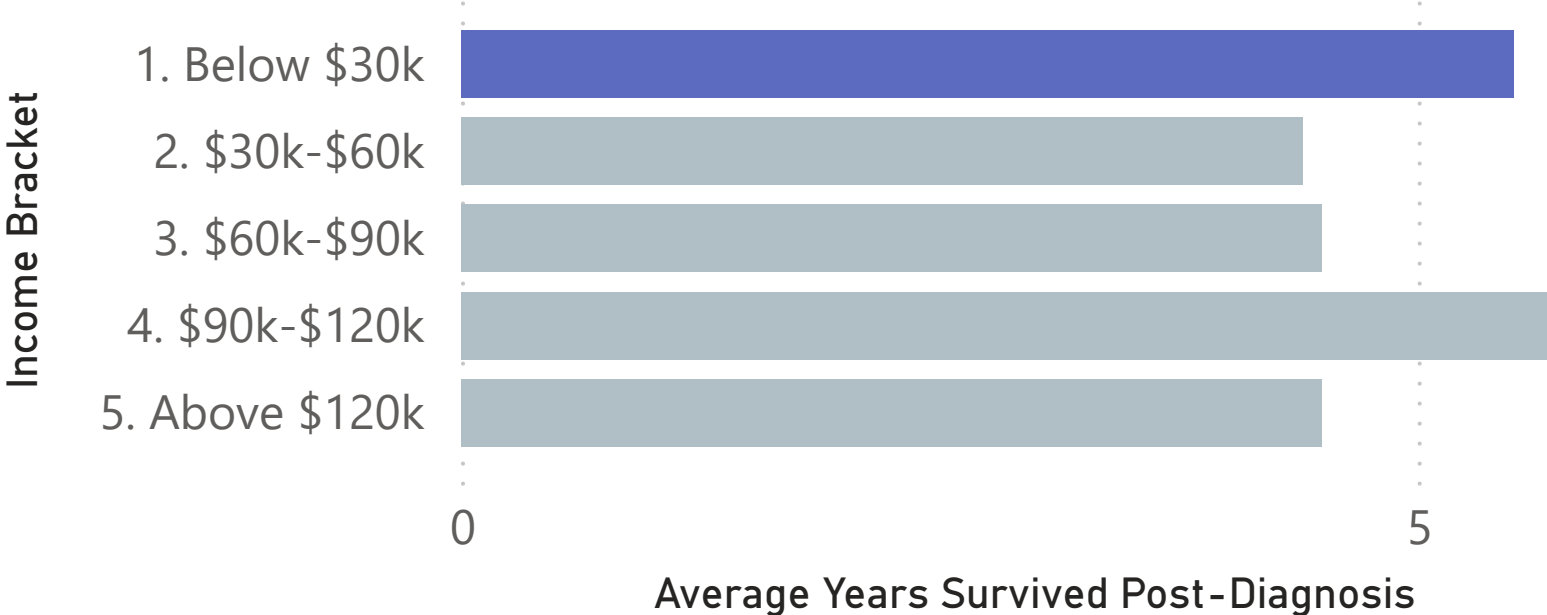
Financial Burden of Cancer Treatment by Income



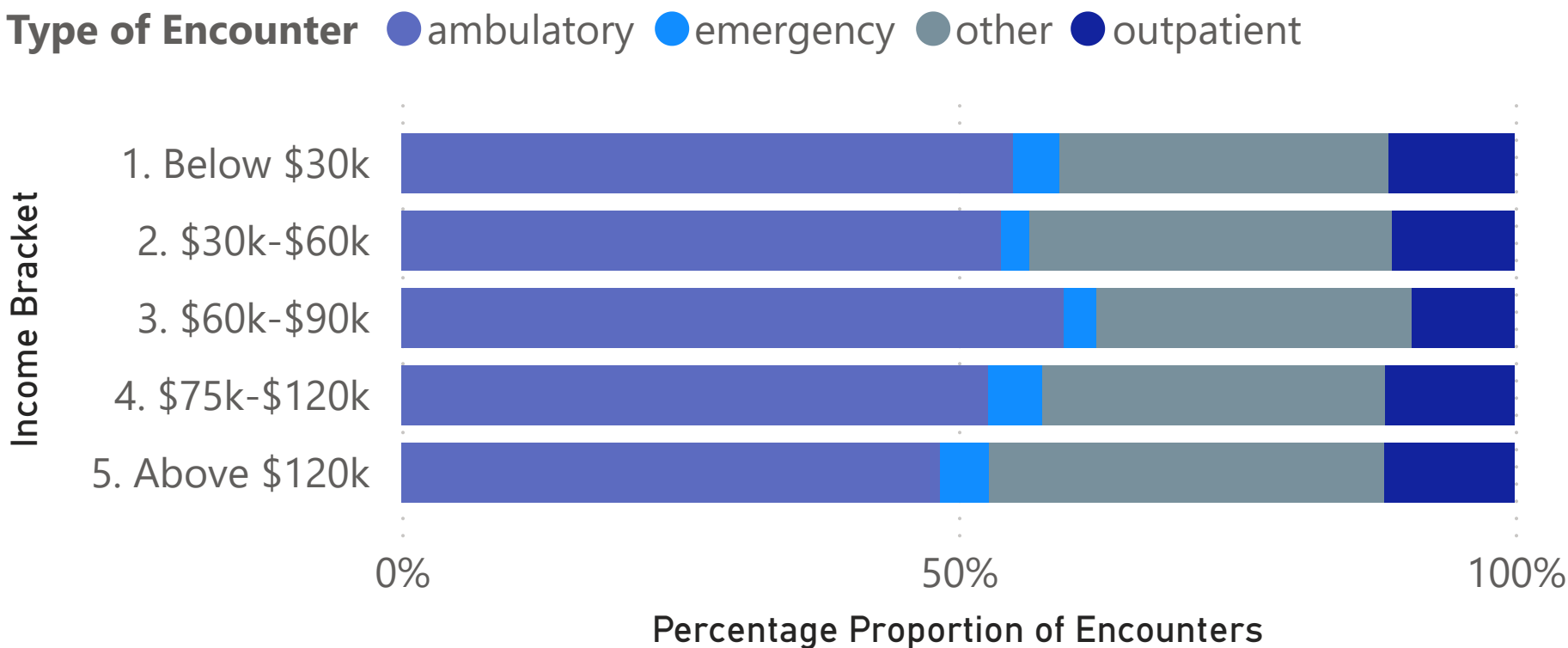
Days from Cancer Diagnosis to First Clinical Encounter



Survival After Cancer Diagnosis by Income



Relationship Between Encounter Type and Income Bracket



Analysis of 10,000 synthetic patient records generated using Synth indicates that individuals in the lowest income bracket (below \$30k) demonstrate the highest cancer diagnosis rates of any income group. Financial burden data reveals that out of pocket treatment costs represent a disproportionately high percentage of income for these patients, creating a significant barrier to timely healthcare engagement.

This financial barrier appears to manifest itself in two measurable ways: a higher average age at cancer diagnosis and a greater number of days between diagnosis and first clinical encounter, indicating delayed diagnosis and treatment.

Notably, survival time post-diagnosis does not follow the expected pattern. Patients in lower income brackets demonstrate comparable or higher average years survived post diagnosis than higher income groups. This may be a result of variations in cancer type distribution or survivorship bias.

Recommendations: reduce financial barriers to care and create early cancer screening initiatives.

Further research: Cancer type distribution across income brackets as well as an analysis of undiagnosed cancer statistics.